engineering 4 us all

S Student Handout

Name: ______________________ **Date:** ______________ **Class:** ____________

MyDesign Portfolio

Element G Initial Template

Details about all steps of construction (**add step numbers as needed**), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Saw a pvc pipe in 3 equal-ish pieces	Jigsaw and clamp
Step 2	Use a drill to add a different amount of holes to each pipe	Power drill
Step 3	Use a dremel to remove excess plastic and round the holes	Handheld dremel
Step 4	Drill 2 small holes next to each other on the other side of the big holes, repeat for both ends of pvc pipe, leaving and inch~ between ends and holes	Power drill
Step 5	Thread wire through one small hole and out the one next to it	Wire

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Step 6	Adjust the wire length depending on desired angle	plyers
Step 7	Hang on ceiling/wall fixture	Hooks, ladder/tall chair

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

STOP

STOP HERE—Create the initial prototype and test it. After you have iterated on the initial design, you will complete the second part of this document. You may iterate more than once.

Final Prototype Construction of Design #3

Date:
Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:
Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used

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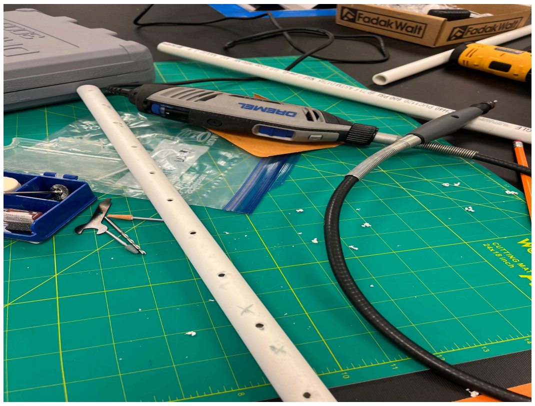
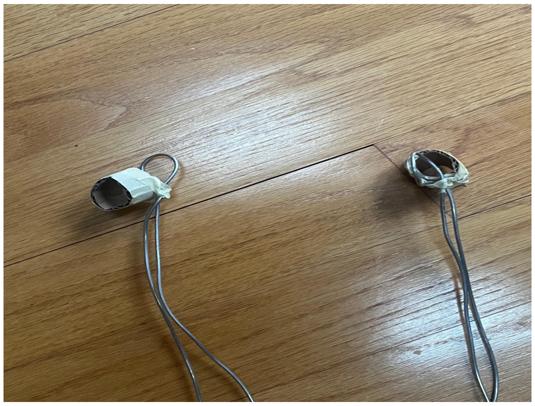
Step 1	Drill holes in the pipe on the same side(5-7 holes) Use a dremel to clean them	Drill, dremel 
Step 2	Take a ½ - 1 inch strip of thin cardboard and create a tight fitting loop, cut down to size and tape the ends together. Slide the ring onto one end of the pipe, a decent way from the edge, make sure it is not covering any holes.	Cardboard, knife, tape 
Step 3	Take aluminum wire and fold in half around the pipe, tape to cardboard ring(repeat steps 2&3 on the other side of pipe)	Wire, tape
Step 4	Attach a sprinkler to the bottom of each	Thin plastic, scissors, glue

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	hole(a plastic star shape with little pillars on on side to glue to holes)	 

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Step 5	Adjust length of wires(bend wires shorter/longer) depending on the wanted angle	
Step 6	Add pipe-screw-attachment thing to higher end	Pipe-screw-attachment thing
Step 7	Bend wires around ceiling fixture to test	Ceiling fixture(shower pole, ceiling hooks, etc.)

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	**Description**	**Where material was purchased**	**Cost per unit**	**# units**	**Total cost per item**
PVC pipe	White pipe, plastic	School	free.99	1	0.00
Aluminum wire	Flexible wire made of aluminum	School	N/A	2	0.00
Pipe-screw-attachment thing	White, plastic, has	School	N/a	1	0.00

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	reverse screw				
Scrap cardboard	Thin, brown	School	N/A	1	0.00
Plastic sheet	Anycolor, thin	Recycling bin	N/A	1	0.00

Detailed sketches, which might include CAD if you have learned it in your class:

Construction of the Final Prototype *(G2)*

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Ease of use	By having a completed prototype we will be able to ask people how they think it should be operated/installed
Mold-lessness	We will be able to run water through it and assess how much mold grows and how quickly it does so.
Distributes water	It will allow us to see how far the water spreads and the distribution of the water, concentrated, spread out, even, uneven.

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Testing or Modeling of Attributes *(G3, G4)* and Addressing Non-Testable Attributes *(G4)*

Attribute	**Description**	**Describe your prototype's functionality to test this attribute**
Mold-less	No mold build up is allowed in the pipe or sprinkler	The prototype has the parts that could potentially mold(ie. Pipe and sprinklers) and is able to run water through it like it would in the final.
Easy to use	Most people are able to quickly understand or learn how to use it	Expert review required
Even distribution	The water should be able to spread to a wide area(~8' by 3')	The sprinkler have been switched to have a smaller hole to add pressure and the deflector below

Addressing Non-Testable Attributes *(G4)*

Attribute	**Justification for why this attribute cannot be modeled or tested with your prototype**
Easy to use	Although it has already been tested with strangers/people with no association to the greenhouse, it still hasn't been tested by the greenhouse club or Mr. Doffermyre/ our experts.

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Element G: Construction of a Testable Prototype						
	5	4	3	2	1	0
1. Final Prototype Iteration Explanation. The Final Prototype Iteration is ______________.	clearly and *fully* explained	*clearly and adequately explained*	*adequately explained*	*somewhat explained but with gaps*	*only minimally* explained	is *unclear* or is *missing* altogether
2. Construction of the Final Prototype. The Final Prototype Iteration is ______________.	constructed with enough detail to assure that objective data on *all* or nearly all design requirements could be determined	constructed with enough detail to assure that objective data on *many or nearly all* design requirements could be determined	constructed with enough detail to assure that objective data on *some* design requirements could be determined	constructed with enough detail to assure that objective data on at least a *few* design requirements could be determined	*not* constructed with enough detail to assure that objective data on at least *one* design requirement could be determined	*not applicable* to any design requirements and/or functional claims
3.Testing or Modeling of Attributes. ______________ attributes (sub-systems) of the unique solution that can be tested or modeled mathematically are	*All*	*Most*	*Some*	*A Few*	*No more than one*	*None of the*

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addressed.						
4.Addressing non-Testable Attributes. _____________ justification is provided for those that cannot be tested or modeled mathematically and thus require expert review.	A *well* supported	A *generally* supported	An *adequately* supported	*Developing*	*Insufficient*	*No*